LESSON PLAN · SCRATCH (GR 6–8) ADVANCED: ALGORITHMS LAB

Lesson 6: Project: Live Leaderboard

Module 2 | 50 minutes | Project: sorting app | Grades 6–8 · ages 12–14

Objective: Plan, build and test a leaderboard app that collects scores, sorts them and shows the top three.

Success criteria

- My app collects and sorts scores.
- It announces the top three and handles small groups.
- I tested it with planned test sets.

Key vocabulary

- **leaderboard**: A ranked list of scores.
- **edge case**: An unusual input at the limits.
- **test set**: A planned group of test inputs.

You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)
- Printed project brief

Timed lesson plan (50 min)

Time	Stage	What happens
5 min	**Warm-up (unplugged)**	Sports day (5 min): rank five race times on the board by bubble passes.
5 min	**Watch & plan**	Play the project video; learners plan on the project brief.
25 min	**Build**	Build-along steps, then Plan the app. Check in halfway: everyone has something working.
7 min	**Test & peer feedback**	Swap and test each other's work: two stars and a wish.
8 min	**Improve & submit**	Improve one thing; Project: Live Leaderboard.

Unplugged warm-up

Sports day (5 min): rank five race times on the board by bubble passes.

Teaching the idea

Combines input collection, sorting with procedures, indexed output and edge-case handling, with systematic test sets (sorted, reversed, ties, tiny).

Analogy: A sports results board updated after every match.

Live demo: Run the leaderboard with 2 players to show a missing-3rd bug, then fix it.

Questions to ask

Ask	Listen for
What happens with 2 players?	Must not show a 3rd.
What about ties?	Both kept; order stable.
Why reuse procedures?	Shorter, tested code.

Differentiation

Support
Provide the sort; learners add collection and reporting.

Stretch
Store names in a parallel list and swap them too.

Common misconceptions

Misconception	How to fix it
The list is sorted as you add	Sort after collecting.
There are always 3 players	Handle edge cases.

Assessment

Project rubric (also on the project brief):

Criterion	1 · Beginning	2 · Developing	3 · Secure	4 · Excellent
Sorting	Not sorted	Partly sorted	Correctly sorted descending	Efficient (early stop) and correct
Input & output	Fixed data	Collects scores	Collects and reports top three	Clear, friendly announcements
Edge cases	Breaks	One handled	Small groups handled	Small groups and ties handled
Procedures	None	swap only	swap and sort procedures	Clean modular design
Testing	None	One run	Four planned test sets	Test sets with expected vs actual

Exit ticket
One edge case your app handles.

Homework
Run your leaderboard for a family game and keep the results.

Answer key — Kitabash activities

1. Plan the app (perform · ordering)

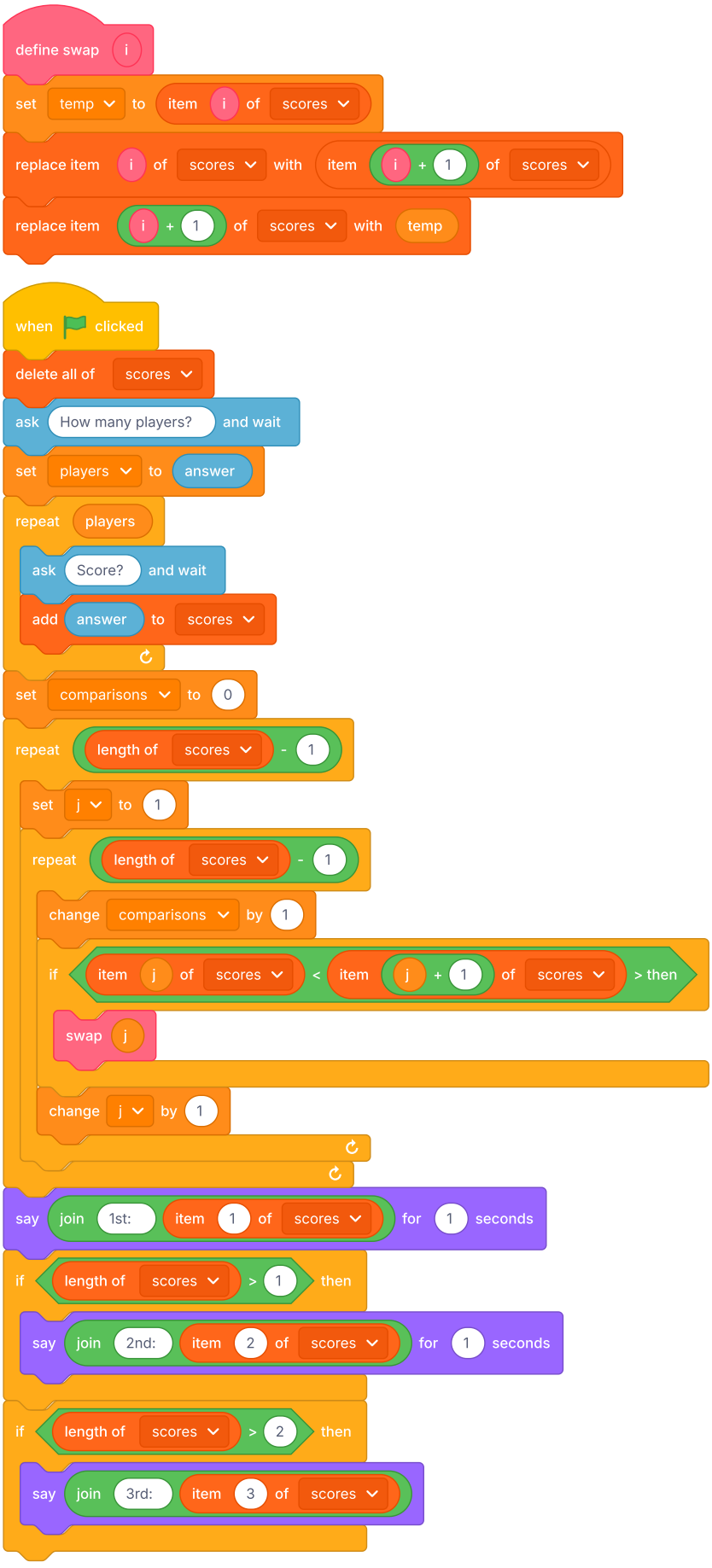
Order: Delete all of scores → Ask how many players → Collect each score into the list → Sort the list descending → Announce the top three

2. Project: Live Leaderboard (probe · code lab)

Goals checked: Collects and sorts · Announces the winner · Uses procedures · Builds messages

Example project:

Sorter


define swap i
set temp to item i of scores
replace item i of scores with item i + 1 of scores
replace item i + 1 of scores with temp
when clicked
delete all of scores
ask How many players? and wait
set players to answer
repeat players
ask Score? and wait
add answer to scores
set comparisons to 0
repeat length of scores - 1
set j to 1
repeat length of scores - 1
change comparisons by 1
if item j of scores < item j + 1 of scores > then
swap j
change j by 1
say join 1st: item 1 of scores for 1 seconds
if length of scores > 1 then
say join 2nd: item 2 of scores for 1 seconds
if length of scores > 2 then
say join 3rd: item 3 of scores

Answer key — worksheet

1. **Key words:** leaderboard = A ranked list of scores. · edge case = An unusual input at the limits. · test set = A planned group of test inputs.
2. **Plan the app:** 1 Delete all of scores · 2 Ask how many players · 3 Collect each score into the list · 4 Sort the list descending · 5 Announce the top three
3. **My plan:** A clear plan that covers every requirement.
4. **Checklist:** All requirements met.
5. **Two stars and a wish:** Kind, specific feedback and an improvement.

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.